

Problem

Determine the Laplace transforms of these functions:

(a) $f(t) = (t - 4)u(t - 2)$

(b) $g(t) = 2e^{-4t}u(t - 1)$

(c) $h(t) = 5 \cos(2t - 1)u(t)$

(d) $p(t) = 6[u(t - 2) - u(t - 4)]$

Step-by-step solution

Step 1 of 6

(a)

Consider the following signal.

$$f(t) = (t - 4)u(t - 2)$$

Apply Laplace transform on both sides.

$$\begin{aligned} F(s) &= \int_{-\infty}^{+\infty} f(t)e^{-st} dt \\ &= \int_{-\infty}^{+\infty} (t - 4)u(t - 2)e^{-st} dt \\ &= \int_{-\infty}^2 (t - 4)u(t - 2)e^{-st} dt + \int_2^{\infty} (t - 4)u(t - 2)e^{-st} dt \end{aligned}$$

It is known that,

$$u(t - 2) = \begin{cases} 0 & t \leq 2 \\ 1 & t > 2 \end{cases}$$

Hence,

$$\begin{aligned} F(s) &= \int_{-\infty}^2 (t - 4)(0)e^{-st} dt + \int_2^{\infty} (t - 4)(1)e^{-st} dt \\ &= 0 + \int_2^{\infty} (t - 4)e^{-st} dt \\ &= \int_2^{\infty} te^{-st} dt - \int_2^{\infty} 4e^{-st} dt \\ &= \int_2^{\infty} te^{-st} dt - \int_2^{\infty} 4e^{-st} dt \end{aligned}$$

Step 2 of 6

Further simplification of yields,

$$\begin{aligned}
 F(s) &= \int_2^{\infty} t e^{-st} dt - \int_2^{\infty} 4 e^{-st} dt \\
 &= \left[\frac{t e^{-st}}{-s} - \left(-\frac{1}{s} \right) \int_2^{\infty} e^{-st} dt \right] + 4 \left(\frac{e^{-st}}{s} \right)_2^{\infty} \\
 &= \left[-\frac{t e^{-st}}{s} - \left(\frac{e^{-st}}{s^2} \right) \right]_2^{\infty} + 4 \left(\frac{e^{-st}}{s} \right)_2^{\infty} \\
 &= - \left[0 - \frac{2e^{-2s}}{s} - \frac{e^{-2s}}{s^2} \right] + 4 \left(\frac{0}{s} - \frac{e^{-2s}}{s} \right) \\
 &= \frac{2e^{-2s}}{s} + \frac{e^{-2s}}{s^2} - \frac{4e^{-2s}}{s} \\
 &= \frac{e^{-2s}}{s^2} - \frac{2e^{-2s}}{s}
 \end{aligned}$$

Therefore, the Laplace transform of $f(t)$ is $\boxed{\frac{e^{-2s}}{s^2} - \frac{2e^{-2s}}{s}}$.

Step 3 of 6

(b)

Consider the following signal.

Apply Laplace transform on both sides.

$$\begin{aligned}
 G(s) &= \int_{-\infty}^{+\infty} f(t) e^{-st} dt \\
 &= \int_{-\infty}^{+\infty} 2e^{-4t} u(t-1) e^{-st} dt \\
 &= \int_{-\infty}^1 2e^{-4t} u(t-1) e^{-st} dt + \int_1^{\infty} 2e^{-4t} u(t-1) e^{-st} dt
 \end{aligned}$$

It is known that,

$$u(t-1) = \begin{cases} 0 & t < 1 \\ 1 & t \geq 1 \end{cases}$$

Hence,

$$\begin{aligned}
 G(s) &= \int_{-\infty}^1 2e^{-4t} (0) e^{-st} dt + \int_1^{\infty} 2e^{-4t} (1) e^{-st} dt \\
 &= 0 + \int_1^{\infty} 2e^{-4t} e^{-st} dt \\
 &= 2 \int_1^{\infty} e^{-(4+s)t} dt \\
 &= 2 \left[\frac{e^{-(4+s)t}}{-(4+s)} \right]_1^{\infty}
 \end{aligned}$$

Step 4 of 6

Simplify the equation further,

$$\begin{aligned} G(s) &= \frac{-2}{4+s} \left[e^{-\infty} - e^{-(4+s)} \right] \\ &= \frac{-2}{4+s} \left[0 - e^{-(4+s)} \right] \\ &= \frac{2e^{-(4+s)}}{4+s} \\ &= \frac{2e^{-s}}{e^4(s+4)} \end{aligned}$$

Therefore, the Laplace transform of $g(t)$ is $\boxed{\frac{2e^{-s}}{e^4(s+4)}}$.

Step 5 of 6

(c)

Consider the following signal.

$$h(t) = 5 \cos(2t-1)u(t)$$

Use the following formula:

$$\text{Since } \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\begin{aligned} h(t) &= \left[5 \cos(2t) \cos 1 + \cos(2t) \sin 1 \right] u(t) \\ &= 5 \cos 1 \cos(2t) u(t) + 5 \sin 1 \sin(2t) \end{aligned}$$

It is known that,

$$\cos(at) \Leftrightarrow \frac{s}{s^2 + a^2}$$

$$\sin(at) \Leftrightarrow \frac{a}{s^2 + a^2}$$

Apply Laplace transform on both sides.

$$\begin{aligned} H(s) &= 5 \cos 1 \left(\frac{s}{s^2 + 2^2} \right) + 5 \sin 1 \left(\frac{2}{s^2 + 2^2} \right) \\ &= \frac{2.702s}{s^2 + 4} + \frac{8.415}{s^2 + 4} \end{aligned}$$

Therefore, the Laplace transform of $h(t)$ is $\boxed{\frac{2.702s}{s^2 + 4} + \frac{8.415}{s^2 + 4}}$.

Step 6 of 6

(d)

Consider the following signal.

$$p(t) = 6[u(t-2) - u(t-4)]$$

Apply Laplace transform on both sides.

$$\begin{aligned} P(s) &= \int_{-\infty}^{+\infty} f(t) e^{-st} dt \\ &= \int_{-\infty}^{+\infty} 6[u(t-2) - u(t-4)] e^{-st} dt \\ &= 6 \int_{-\infty}^{+\infty} [u(t-2) - u(t-4)] e^{-st} dt \end{aligned}$$

It is known that,

$$u(t-2) - u(t-4) = \begin{cases} 1 & 2 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Hence,

$$\begin{aligned} P(s) &= 6 \int_2^4 [1] e^{-st} dt \\ &= 6 \left[\frac{e^{-st}}{-s} \right]_2^4 \\ &= -\frac{6}{s} [e^{-4s} - e^{-2s}] \\ &= \frac{6}{s} e^{-2s} - \frac{6}{s} e^{-4s} \end{aligned}$$

Therefore, the Laplace transform of $p(t)$ is $\boxed{\frac{6}{s} e^{-2s} - \frac{6}{s} e^{-4s}}$.